Weather, Mental-Health Treatment, and Burnout Prediction

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STAT 390 Final Presentation

Question: do weather variables add predictive value beyond work, sleep, health, and demographic variables?

Research Question

- Prior research suggests weather can affect mood, depression, and performance.
- This project tests a narrower predictive question.
- Does weather improve model performance after stronger personal and workplace predictors are already included?
- Targets: sought_treatment and PCA-derived burnout_index.

Data and Pipeline

- Mental-health survey: treatment target and workplace mental-health predictors.

- Sleep-health dataset: sleep, stress, work, and health variables.

- Weather data: temperature, wind, pressure, sunlight, humidity, and related features.

- Burnout index: PCA target built from non-weather stress/sleep/work indicators.

- Weather was excluded from index construction to avoid leakage.

Controlled Experiment Design

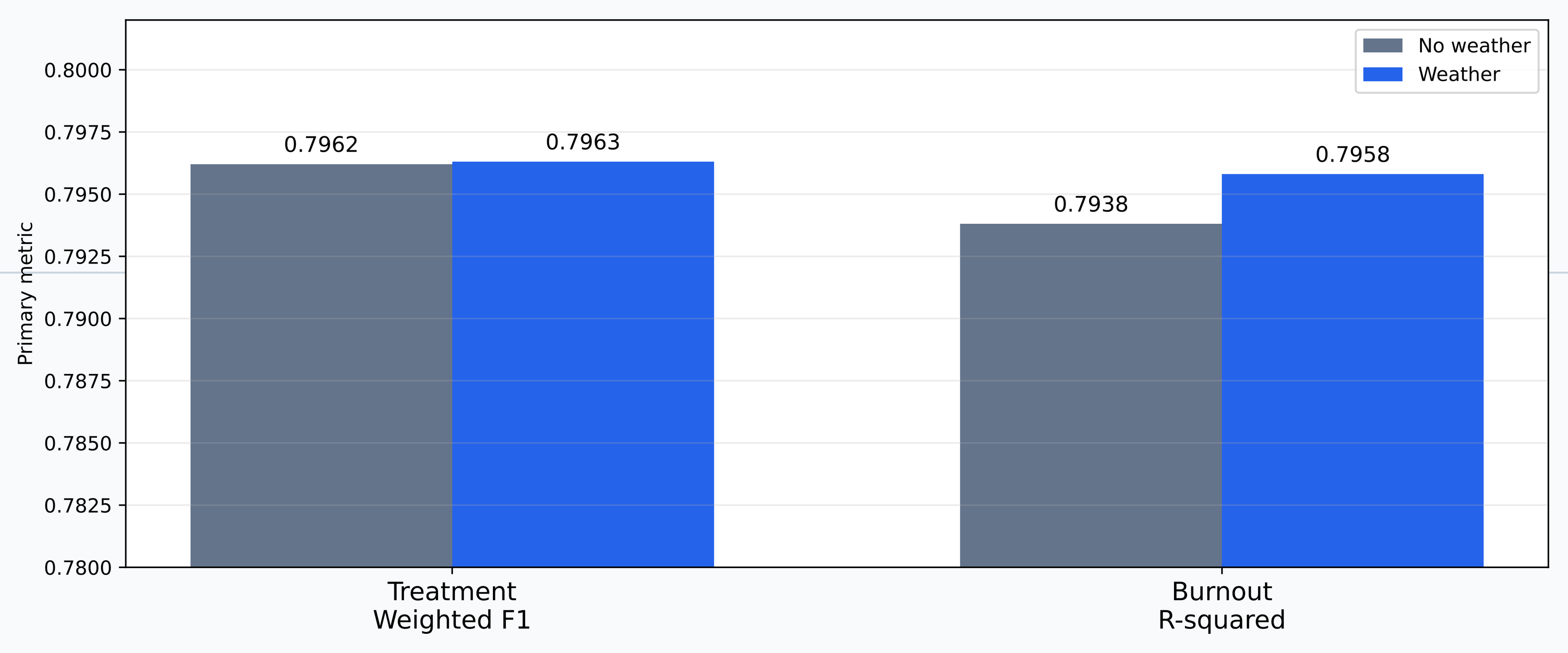
- For each target, compare non-weather features vs. weather-augmented features.
- Same random seed, split policy, preprocessing, and model list inside each comparison.
- Classification metric: weighted F1.
- Regression metric: R-squared.
- Full run: 88 completed models, 0 failures.

Baseline vs. Best Model

Outcome	Baseline	Best	Gain
Treatment, no weather	0.3915	0.7962	+0.4047
Treatment, weather	0.3915	0.7963	+0.4048
Burnout, no weather	~0.0000	0.7938	+0.7939
Burnout, weather	~0.0000	0.7958	+0.7959

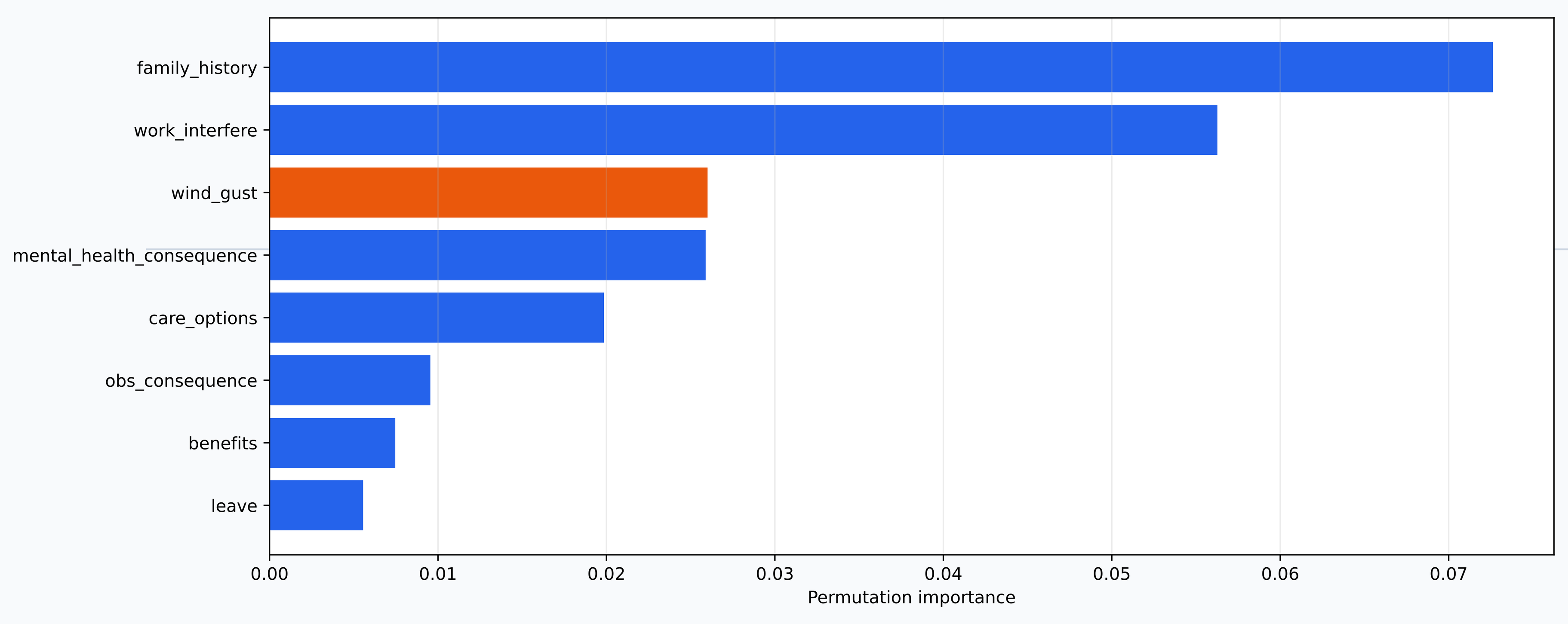
Main point: the models learned real predictive structure, even though weather added only a small extra gain.

Weather vs. Non-Weather Results



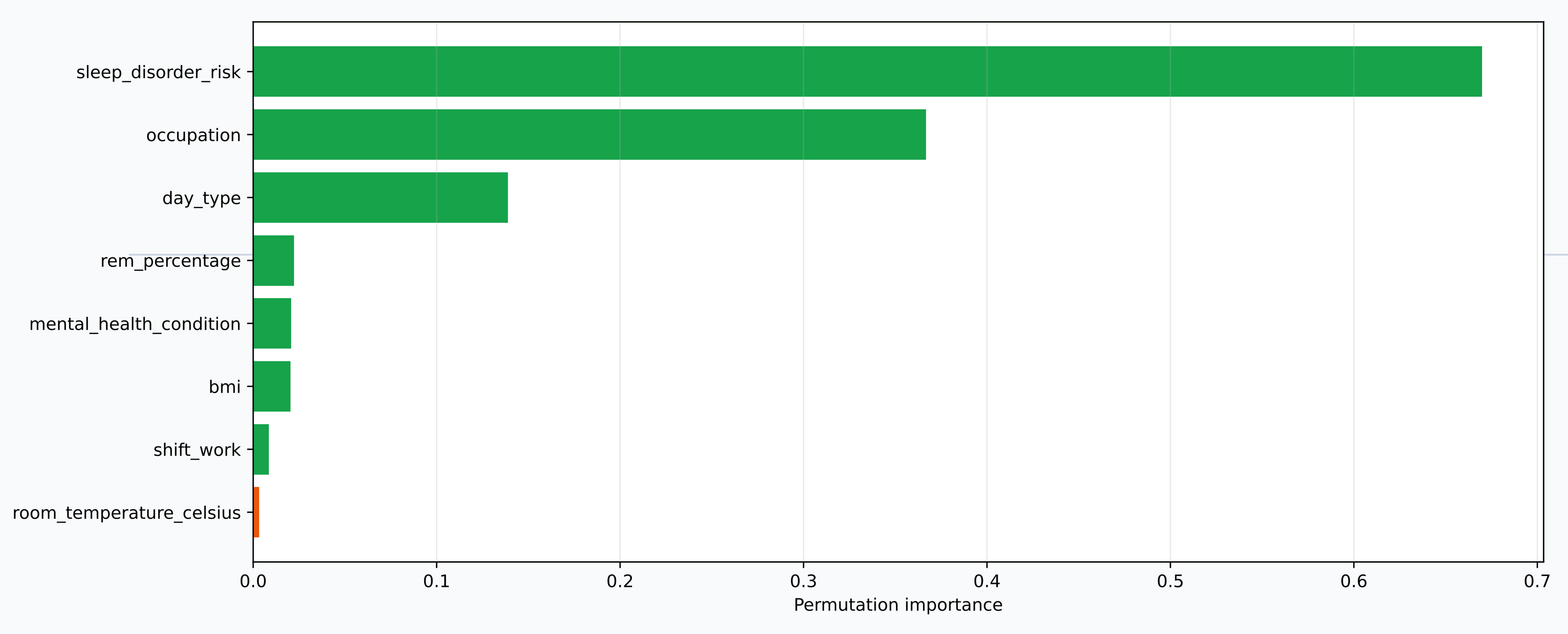
Weather improved both controlled runs, but the gains were small.

Treatment Prediction: Weather Signal



Weather did not change F1 much, but wind_gust was the third most important treatment feature.

Burnout Index: Weather Is Weaker



Burnout prediction is dominated by sleep, occupation, and day-type variables.

Interpretation

- The project does not disprove prior weather-and-mental-health research.
- It asks whether weather adds prediction after stronger variables are already included.
- Weather effects may be indirect through sleep, stress, work patterns, or treatment behavior.
- Treatment prediction showed the clearest weather-related importance signal.
- Burnout index prediction was mostly explained by direct sleep/work/health variables.

Limitations and Next Steps

- Weather matching is coarse and may not represent each person's true exposure.

- Timing may matter: heat waves, recent sunlight, seasonal lag, or multi-day exposure.

- The burnout index is constructed from variables close to sleep and work stress.

- Future work: finer geographic weather, date-aligned exposure windows, and causal controls.

Final Conclusion

- Models improved strongly over dummy baselines.

- Weather added weak incremental predictive value overall.

- Treatment prediction had the clearest weather importance signal: wind_gust and pressure_hpa.

- Burnout prediction was driven more by sleep, occupation, day type, and health variables.

- Final claim: weather may matter indirectly, but it is not a major predictor in this dataset.